

1. The Problem

HealthConnect is trying to reduce **patient appointment no-shows** and improve clinic attendance and planning.

2. The Data

We used **appointments, patients, booking, reminder, distance, waiting-time and previous attendance data** from the HealthConnect dataset.

3. Analytics

The analysis showed that **48.23% of appointments were no-shows**, with distance, reminders, age and previous attendance showing useful patterns.

4. The Predictive Component

Data Science was used to build a **machine learning model that predicts whether a patient is likely to attend or not show up**.

5. The ML Engineering Component

The model was organized into a workflow covering **data preparation, feature engineering, model training, testing, prediction and saving the model for application use**.

6. The GenAI Component

The AI assistant can support healthcare administration by **helping staff access information, answer routine questions and improve the patient's communication experience**.

7. The Project Management Layer

The project was divided into different workstreams, with **clear tasks, weekly deliverables, testing and refinement** to bring the components together.

8. The Final Solution

The **Data Analysis Insights predictive model**, work together to provide HealthConnect with a more data-driven approach to managing appointments.

9. Business Value

The solution could help HealthConnect **identify high-risk no-shows, improve reminder strategies, plan resources and support better appointment attendance**.

10. Limitations

The model currently depends on the **quality and availability of historical data** and cannot account for unexpected circumstances that may cause a patient to miss an appointment.